



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SESHASAYEE PAPER AND BOARDS LTD-RO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

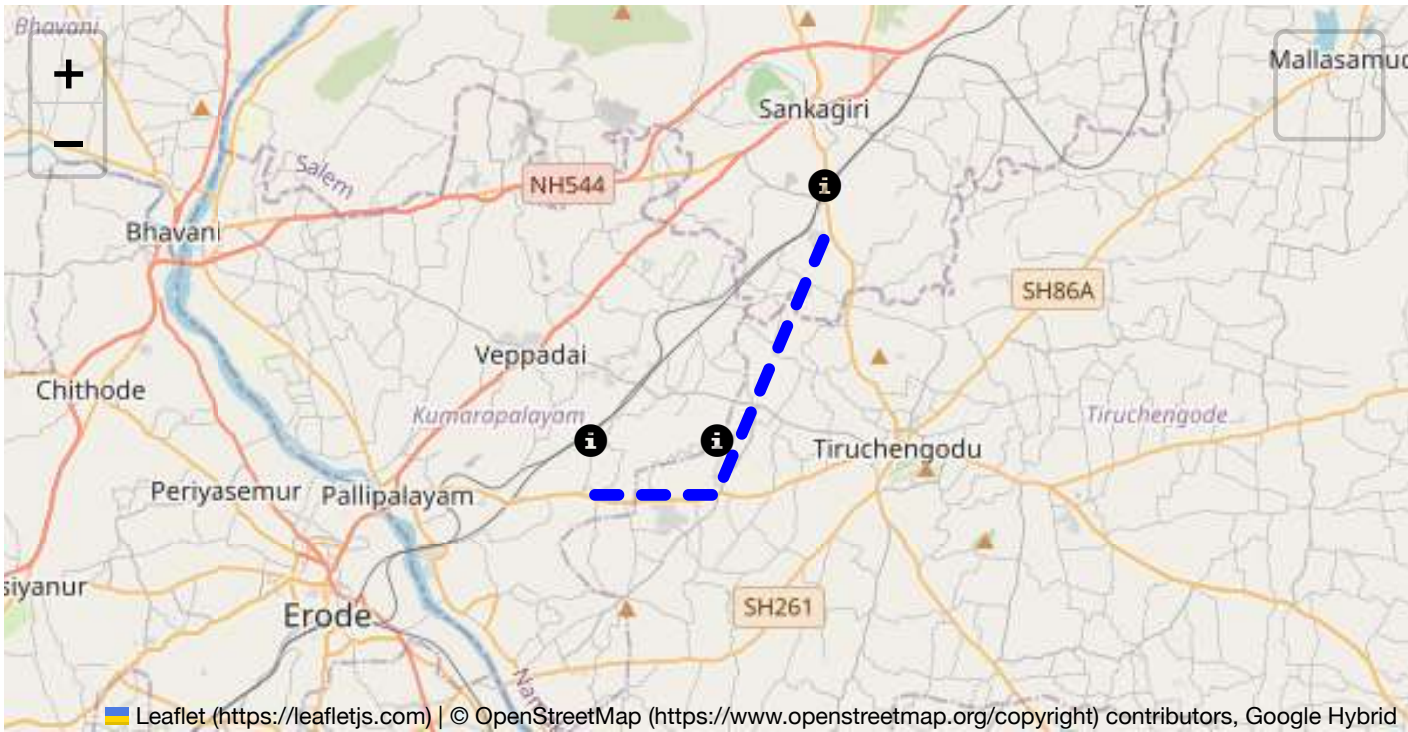
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

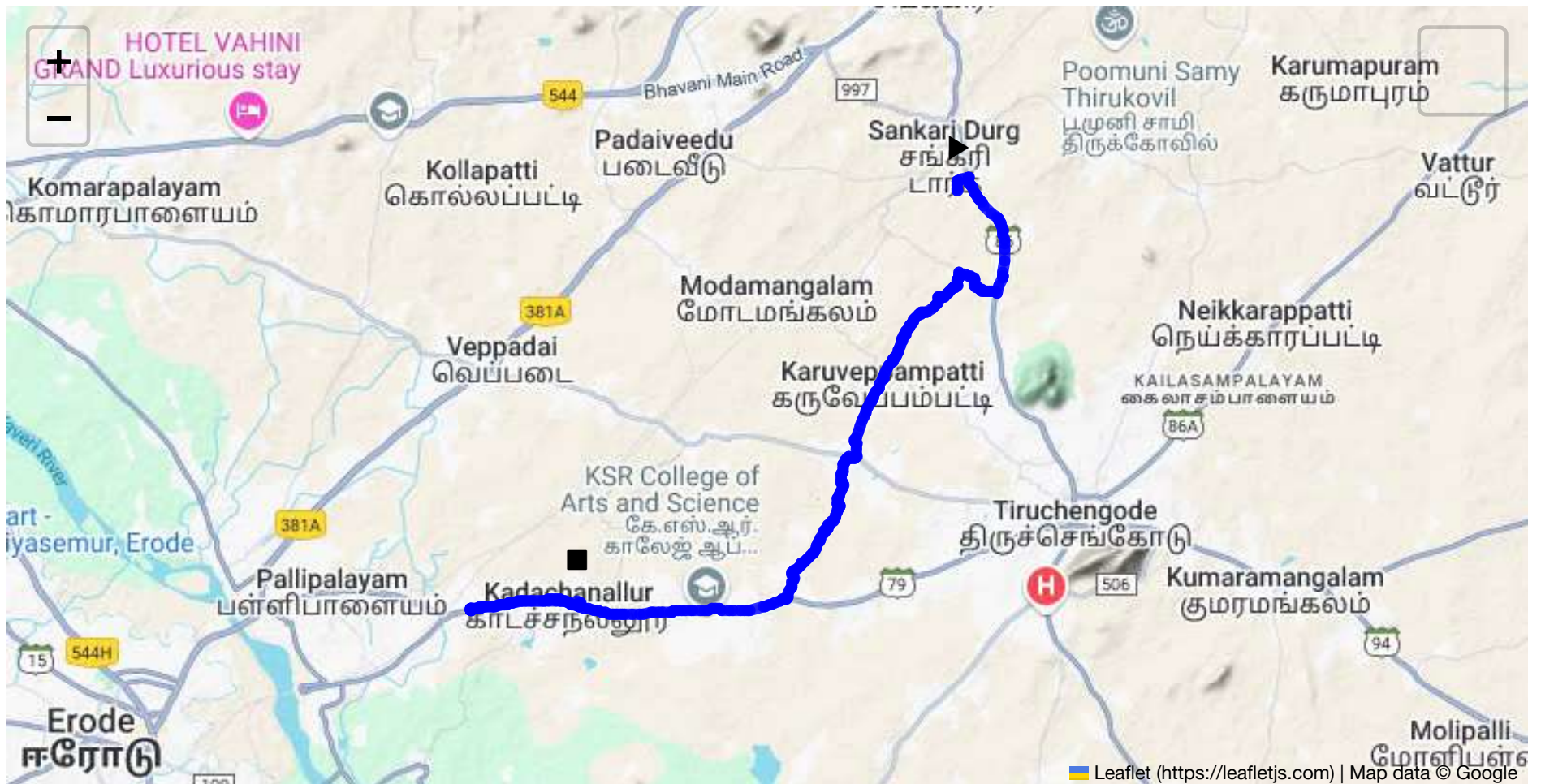


Route Summary:
Total Distance: 20.38 km
Estimated Duration: 0.6 hours

Adjusted Duration (Heavy Vehicle): 0.7 hours

Start: (11.4381, 77.8734)

End: (11.3649, 77.8046)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route approximately spans 20.38 kilometers from Sangagiri to Aindhupanai, passing through Thiruchengodu. The journey involves navigating through both rural and semi-urban areas, predominantly along state highways that connect these towns. The terrain is mostly flat, with sections of agricultural lands and small settlements.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. Key weather patterns include:

- **Summer (March to June):** High temperatures reaching up to 40°C which can cause road surfaces to soften, potentially affecting vehicle grip.
- **Monsoon (July to September):** Heavy rainfall leading to waterlogging, especially in low-lying areas, which might affect visibility and road conditions.
- **Winter (October to February):** Milder temperatures; however, occasional fog during early mornings could hinder visibility.

No extreme weather conditions like snow or ice are prevalent, but sudden showers during the monsoon need attention.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic tends to be heavier during the morning commute from 7:30 AM to 9:30 AM and evening return from 5:30 PM to 7:30 PM.

- **Congestion-Prone Areas:** Key congestion points are likely near market areas in Thiruchengodu due to commercial activity, especially during weekends.
- **Truck Restrictions:** Be mindful of certain roads in Thiruchengodu that may not permit heavy vehicles during peak hours.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Generally, the state highways are well-maintained, but expect occasional potholes, especially after the monsoon. Rural sections might have narrower lanes.
- **Infrastructure:** Adequate signage is present, but lighting may be deficient in remote segments.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route 1:** Route via NH44, which offers better road quality and emergency services but is longer.
- **Alternate Route 2:** Local roads parallel to the primary route can be used, keeping in mind that infrastructure might be less developed.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is regulated by guidelines requiring:

- Proper labeling and containment.
- Restrictions on movement times, often limiting transport to non-peak hours.
- Special permits may be needed, particularly for navigating Thiruchengodu's populated areas.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Incidents:** There have been occasional reports of truck overturns due to narrow lanes or driver fatigue, but no major hazardous material spills recently.
- **Lessons Learned:** Emphasize training on handling vehicle dynamics on varied road conditions and implementing strict adherence to driving hours to avoid fatigue.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** The route passes near agricultural fields, and any spillage could impact soil and water quality.
- **Environmental Measures:** Ensure that all hazardous materials are adequately secured and drivers are trained to manage spills.

9. Analysis of Communication Coverage

- **Strong Coverage:** In and around Sangagiri and Thiruchengodu.
- **Potential Dead Zones:** Might experience weak signals in certain rural patches; drivers should have updated maps and emergency contacts saved offline.

10. Estimated Emergency Response Times for Different Route Segments

- Urban Areas (Thiruchengodu): 15-30 minutes, given proximity to local hospitals and police.
- Rural Segments: Up to 45 minutes due to distance from emergency facilities.

11. An Overall Summary of Risk Assessment

The route from Sangagiri to Aindhupanai involving Thiruchengodu, while relatively short, presents moderate challenges predominantly due to weather conditions, traffic congestion, and road quality variations. The primary risks involve potential delays and accidents, particularly in monsoon conditions and areas of traffic congestion. Regular vehicle maintenance and adherence to local transport guidelines will mitigate most risks. Training for all drivers on handling hazardous materials and emergency situations effectively will further enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.36373, 77.78567	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	Medium	11.43971, 77.87348	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Blind Spot	Blind Spot	11.41981, 77.88078	10 KM/Hr
6	Turn	Medium	11.42102, 77.87680	30 KM/Hr
7	Turn	Medium	11.42231, 77.87666	30 KM/Hr
8	Blind Spot	Blind Spot	11.42333, 77.87394	10 KM/Hr
9	Turn	Medium	11.42129, 77.87359	30 KM/Hr
10	Turn	Medium	11.42090, 77.87311	30 KM/Hr
11	Turn	High	11.41917, 77.87022	15 KM/Hr
12	Turn	Medium	11.41642, 77.86800	30 KM/Hr
13	Turn	High	11.40478, 77.86020	15 KM/Hr
14	Turn	High	11.40471, 77.86004	15 KM/Hr
15	Turn	Medium	11.39360, 77.85502	30 KM/Hr
16	Turn	Medium	11.39325, 77.85538	30 KM/Hr
17	Turn	High	11.39047, 77.85509	15 KM/Hr
18	Turn	High	11.39076, 77.85379	15 KM/Hr
19	Turn	Medium	11.38629, 77.85258	30 KM/Hr
20	Turn	High	11.38255, 77.85172	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
21	Turn	Medium	11.38244, 77.85179	30 KM/Hr
22	Turn	Medium	11.38221, 77.85219	30 KM/Hr
23	Turn	Medium	11.38189, 77.85225	30 KM/Hr
24	Turn	Medium	11.37019, 77.84376	30 KM/Hr
25	Turn	High	11.36568, 77.84272	15 KM/Hr
26	Turn	High	11.36438, 77.80523	15 KM/Hr
27	Blind Spot	Blind Spot	11.36494, 77.80546	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Government Hospital Kadachanallur	11.36702, 77.804315	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	K. S. Rangasamy College of Technology	11.3587246, 77.8278764	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36373, 77.78567



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41981, 77.88078



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42333, 77.87394



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41917, 77.87022



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41642, 77.86800



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40478, 77.86020



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40471, 77.86004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39360, 77.85502



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39325, 77.85538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39047, 77.85509



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39076, 77.85379



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38629, 77.85258



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38255, 77.85172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38244, 77.85179



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38221, 77.85219



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38189, 77.85225



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37019, 77.84376



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.36568, 77.84272



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.36438, 77.80523



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36494, 77.80546

